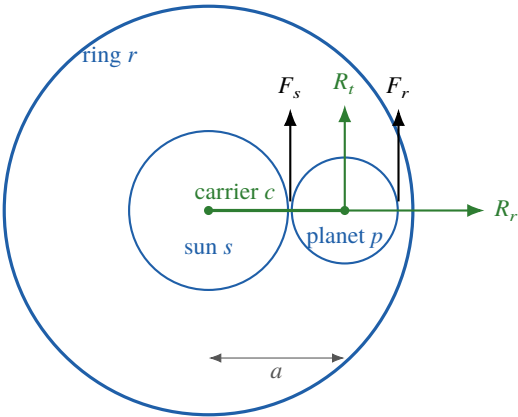




external shafts:

$$\tau_s(\omega_s - \Omega)$$

$$\tau_r(\omega_r - \Omega)$$

$$\tau_c(\omega_c - \Omega)$$

one planet boundary:

two contact sides

one pin side

independent spin/orbit store